BYU CAPSTONE TEAM 33

Capstone User Guide

End-user guides for Cline, RAG Website, and OpenWebUI

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USER GUIDE

WireGuard User Guide

Overview

WireGuard is the VPN used to connect your computer to the private server that hosts the AI tools (OpenWebUI, RAG Website, Cline). All of those services are only reachable over the VPN — they are not exposed to the public internet. You need WireGuard running before you can access any of them.

This guide covers everything a new user needs to do to get connected. The server-side setup is handled by your administrator.

What You Need from Your Administrator

Before you can connect, your administrator needs to add your device to the VPN server. To do that, they need your **public key**, which WireGuard generates for you (covered in Step 2 below).

Your administrator will give you:

- Your assigned **VPN IP address** (e.g. `10.55.55.2`)
- The server's **public key**
- The server's **public IP address**

You will need all three to complete your tunnel configuration.

Step 1: Install WireGuard

Download and install WireGuard for Windows from the official installer:

https://www.wireguard.com/install

Run the installer and open the WireGuard app when it finishes.

Step 2: Create a Tunnel and Get Your Public Key

1. In the WireGuard app, click **Add Tunnel** → **Add empty tunnel**.
2. WireGuard automatically generates a **private key** (already filled in) and a **public key** shown at the top of the editor.
3. Copy your **public key** and send it to your administrator.

> **Keep your private key private**
>
> Never share the private key. Only share the public key with your administrator.

> **Wait before activating**
>
> Do not activate the tunnel yet. Wait until your administrator confirms your public key has been added to the server.

Step 3: Configure the Tunnel

Once your administrator confirms you have been added, replace the contents of the tunnel editor with the following and fill in the three placeholders:

```
[Interface]
PrivateKey = <YOUR_PRIVATE_KEY>
Address = <YOUR_VPN_IP>/32
DNS = 8.8.8.8

[Peer]
PublicKey = <SERVER_PUBLIC_KEY>
Endpoint = <SERVER_PUBLIC_IP>:51820
AllowedIPs = 10.55.55.0/24
PersistentKeepalive = 25
```

- `<YOUR_PRIVATE_KEY>` — already present in the editor from Step 2, leave it as-is
- `<YOUR_VPN_IP>` — the IP address your administrator assigned to you (e.g. `10.55.55.2`)
- `<SERVER_PUBLIC_KEY>` — provided by your administrator
- `<SERVER_PUBLIC_IP>` — provided by your administrator

Click **Save**.

Step 4: Activate the Tunnel

Click **Activate** in the WireGuard app. The status should change to **Active**.

To verify the connection, open a terminal and ping the VPN gateway:

```
ping 10.55.55.1
```

If you get replies, you are connected. You can now open a browser and access the internal services.

Connecting and Disconnecting

WireGuard does not need to run all the time. You can toggle it on and off as needed:

- **To connect:** Open WireGuard and click **Activate** on your tunnel.
- **To disconnect:** Open WireGuard and click **Deactivate**.

You must have the tunnel **Active** to reach any of the internal services.

Internal Service URLs

Once WireGuard is active, open these URLs in your browser:

Service	URL
OpenWebUI (AI chat)	`http://10.55.55.1:3000`
RAG Website (index manager)	`http://10.55.55.1:7000`
LiteLLM	`http://10.55.55.1:4000`
MCP Docs	`http://10.55.55.1:8001/docs`

Troubleshooting

The tunnel activates but I can't reach the services. Make sure the tunnel status shows **Active**, then try pinging `10.55.55.1` in a terminal. If the ping fails, your administrator may need to restart the VPN server or check your peer entry.

WireGuard says "Activation failed". Double-check that the `PublicKey`, `Endpoint`, and `Address` values in your tunnel configuration exactly match what your administrator provided.

I got a new computer and need to reconnect. You will need to repeat Steps 1–4 on the new machine. WireGuard generates a new key pair per installation, so your administrator will need your new public key before you can connect.

USER GUIDE

Cline User Guide

Overview

Cline is an AI-powered coding assistant built into Visual Studio Code. It can read and edit files in your workspace, run terminal commands, and call external tools via MCP servers — including the Azure AI Search indexes you set up through the RAG Website. This guide covers both the initial setup a new user needs to complete and the features available for everyday use.

Setup

Complete these steps once before using Cline for the first time.

Prerequisites

- WireGuard is connected on your local machine

Step 1: Install Visual Studio Code

1. Go to https://code.visualstudio.com/.
2. Download and install the version for your operating system.
3. Launch Visual Studio Code.

Step 2: Install the Cline Extension

1. Open the **Extensions** view with `Ctrl+Shift+X`.
2. Search for **Cline**.
3. Click **Install**.

Step 3: Configure the API Connection

If this is your first time installing Cline, the setup wizard opens automatically:

1. On the "How will you use Cline?" screen, select **Bring my own API key** and click **Continue**.
2. On the "Configure your provider" screen, set:
   - **API Provider:** `OpenAI Compatible`
   - **Base URL:** `http://10.55.55.1:4000`
   - **OpenAI Compatible API Key:** `none`
   - **Model ID:** `gpt-4o`

   Cline API Config

3. Click **Continue**.

If Cline is already installed, open Settings directly:

1. Open the Cline panel from the Activity Bar (left sidebar).
2. Click the **gear icon** in the top-right corner of the Cline panel.
3. Under **API Configuration**, set the same four fields listed above.
4. Click **Done**.

Step 4: Connect to the MCP Server

1. Click the **MCP Servers icon** (plug icon) in the top-right corner of the Cline panel.
2. Open the **Remote Servers** tab.
3. Add a new server with:
   - **Server Name:** `Azure MCP` (or any label you prefer)
   - **Server URL:** `http://10.55.55.1:8000/mcp`
   - **Transport Type:** `Streamable HTTP`

   Cline MCP Config

4. Save the entry.

Step 5: Verify the Connections

Model connection: Open a new Cline chat. The bottom bar of the panel should show the model name (e.g., `openai-compat:gpt-4o`). Send a short message — if Cline responds, the connection is working.

MCP connection: Open MCP Servers, go to the **Configure** tab, and confirm your server entry shows a **green dot**, indicating it is connected.

The Cline Interface

Main Panel

When you open Cline, the panel shows recent conversations at the top and the chat input at the bottom. Type your task in the input field and press **Enter** to send.

The bottom bar of the panel always shows the active model name. The **Plan** and **Act** buttons to the right of it switch between the two operating modes (described below).

Toolbar Icons (Bottom Left)

The four icons to the left of the chat input provide quick access to common actions:

Cline Toolbar

Button	Action
@	**Add Context** — attach files, folders, URLs, or other references to your message
+	**Add Files & Images** — opens the file explorer to attach local files or screenshots
MCP servers icon	**Manage MCP Servers** — view and configure connected MCP tool servers
Rules icon	**Manage Cline Rules & Workflows** — open the panel for Rules, Workflows, Hooks, and Skills

Adding Context

Cline works best when it knows what you are working on. You can add context in several ways.

@ Context Picker

Type `@` in the chat input or click the @ button to open the context picker menu. The following options are available:

Cline @ list

- **Paste URL to fetch contents** — fetches and attaches the contents of a web page
- **Problems** — attaches the current errors and warnings from the VS Code Problems panel
- **Terminal** — attaches the output from the current terminal session
- **Git Commits** — attaches recent git commit history
- **Add Folder** — browse and attach a folder from your workspace
- **Add File** — browse and attach a specific file

Files and Images

Click the **+** icon to open the file explorer and attach local files or screenshots directly to your message. This is useful for sharing error screenshots, diagrams, or files outside your current workspace.

Drag and Drop

Drag a file from the VS Code Explorer directly into the Cline chat input to attach it.

Plan and Act Modes

Cline Plan/Act Config

Cline operates in two modes, toggled with the **Plan** and **Act** buttons at the bottom of the panel.

Plan Mode

In Plan mode, Cline thinks through the task and proposes an approach before doing anything. Use this when:

- The task is complex or has multiple steps
- You want to review the strategy before Cline starts making changes
- You are using a strong reasoning model to design a plan for a cheaper model to execute

Cline will ask clarifying questions and outline what it intends to do. No files are edited and no commands are run until you switch to Act mode.

Act Mode

In Act mode, Cline executes tasks directly. Each file edit or terminal command is shown as a proposed action that you approve or reject before it runs (unless Auto-approve is enabled for that action type).

Use Act mode once you are satisfied with the plan, or for straightforward tasks that do not need a planning step.

Auto-Approve

The **Auto-approve** section at the bottom of the Cline panel controls which actions Cline can take without prompting you each time. This speeds up your workflow for actions you trust.

Cline Auto Approve Config

Permission	What it allows
Read project files	Read files inside the current VS Code workspace
Read all files	Read any file on the machine
Edit project files	Modify files inside the workspace without asking
Execute safe commands	Run low-risk terminal commands (e.g., `ls`, `cat`)
Execute all commands	Run any terminal command without asking
Use the browser	Open and interact with web pages
Use MCP servers	Call MCP tools (e.g., RAG index search) without asking

The default configuration enables **Read project files**, **Execute safe commands**, and **Use MCP servers**. Enable additional permissions only if the interruptions are slowing you down and you trust Cline's judgment for that action type.

> **Warning**
>
> Enabling **Edit project files** or **Execute all commands** means Cline can modify your code and run arbitrary commands without showing you a preview first. Only enable these if you are comfortable with that level of autonomy.

Rules, Workflows, Hooks, and Skills

Click the **Manage Cline Rules & Workflows** icon (bottom left of the chat input) to open the management panel. It has four tabs.

Rules

Rules provide Cline with persistent system-level guidance that applies to every conversation — without you having to repeat yourself each time. Think of them as standing instructions.

Cline Rules

- **Global Rules** apply across all workspaces on your machine.
- **Workspace Rules** apply only to the current VS Code project.

Examples of useful rules: - "Always write comments in English." - "Never modify test files unless I explicitly ask." - "When working in this repo, follow the coding conventions in `CONTRIBUTING.md`."

Click **+** next to the appropriate section and point it at a rule file (a plain text or markdown file containing your instructions).

Workflows

Workflows define a reusable series of steps for repetitive tasks. Once defined, you invoke a workflow by typing `/workflow-name` in the chat.

Cline Workflows

- **Global Workflows** are available in any workspace.
- **Workspace Workflows** are specific to the current project.

Example workflows: - `/deploy` — run tests, build, and push to the server - `/pr` — stage changes, write a commit message, and open a pull request

Hooks

Hooks let you run custom scripts at specific points in Cline's execution lifecycle — for example, automatically running a formatter after every file edit. Hooks are enabled or disabled by setting file permissions (`chmod +x` to enable, `chmod -x` to disable).

Cline Hooks

- **Global Hooks** apply across all workspaces.

- **Workspace Hooks** apply only to the current project (stored in `.clinerules/hooks/`).

Skills

Skills are reusable instruction sets that Cline loads on demand. When a task matches a skill's description, Cline automatically invokes the `use_skill` tool to load the full instructions for that skill.

Cline Skills

- **Global Skills** are available in any workspace.
- **Workspace Skills** are specific to the current project.

Using MCP Tools (RAG Index Access)

When the MCP server is connected and **Use MCP servers** is enabled (either via auto-approve or manual approval), Cline can search your RAG indexes as part of answering a question.

You do not need to invoke tools manually — Cline decides when to use them based on your request. If you want to target a specific index, be explicit:

> "Search the `ipp-documentation` index for information about the API authentication flow."

To see which MCP tools are currently available, click the

icon in the Cline toolbar. Each tool corresponds to one of your configured indexes or other MCP-connected services.

Tips

- **Be specific.** "Fix the bug in `auth.py` where the token expires too early" works better than "fix the bug."

- **Use Plan mode for multi-step tasks.** Let Cline lay out the approach before it starts touching files.
- **Start a new chat when switching tasks.** Context from previous conversations carries over and can confuse Cline on unrelated work.
- **Keep conversations short.** Very long sessions consume more tokens and can degrade response quality. Start fresh if a chat has gone on for a long time.
- **Follow up with corrections.** If an edit is almost right, say so: "That's close, but don't rename the existing variables." Cline will revise.
- **Use workspace rules for project conventions.** Rather than repeating "use 4-space indentation" every session, put it in a workspace rule file.

USER GUIDE

RAG Website User Guide

Overview

The RAG Website is a centralized interface for managing the knowledge bases that power your AI solutions. By uploading documents or scraping web pages, you create **indexes** — isolated collections of content that the AI can search through when answering questions. This guide covers every step of creating, editing, and managing indexes through the Index Manager.

Prerequisites

- Documents you want to upload are ready (PDF, plain text, or source code files)
- WireGuard is connected on your local machine

Key Concepts

What is an Index?

An index is an isolated knowledge base stored in Azure AI Search. Think of it as a dedicated "brain" for a specific topic or project. Each index is independent — the AI only searches the index you point it at, so keeping indexes focused on a single subject improves answer quality.

What are Embeddings and Vector Dimensions?

When a document is added to an index, the system converts each chunk of text into a list of numbers called an **embedding**. An embedding captures the meaning of that text in a way a computer can compare and search — two chunks about similar topics will have embeddings that are numerically close to each other.

Vector Dimensions controls how many numbers are in each embedding. The default is **3072**, which matches the output size of the embedding model in use. You should not change this value unless you are switching to a different

embedding model that produces a different output size, as a mismatch will break the index.

What is Chunk Size?

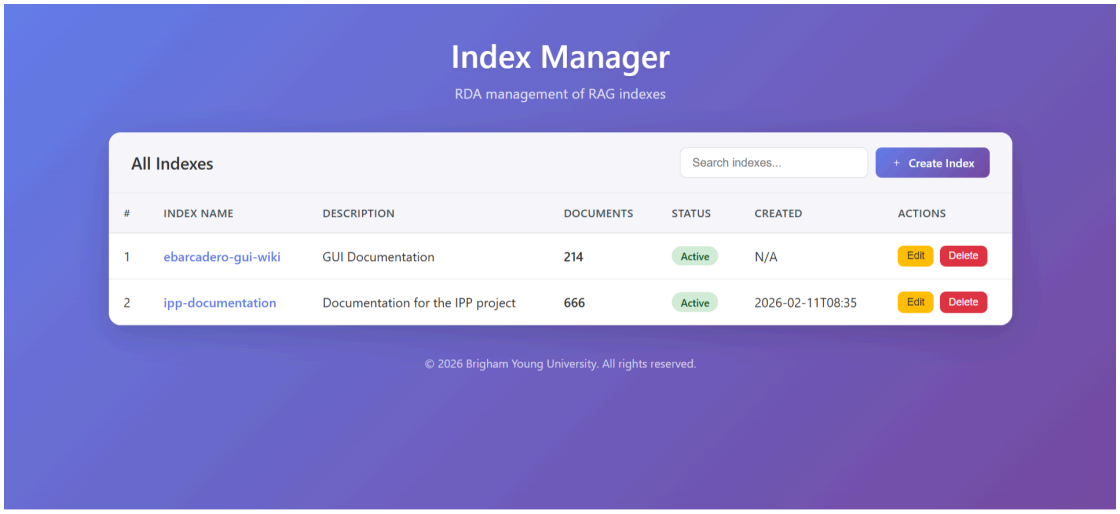
Before a document is indexed, it is split into smaller segments called chunks. **Chunk Size** controls the maximum number of tokens (roughly, word-pieces) in each segment.

- **Smaller chunks** (e.g., 512) give more precise retrieval — the AI gets a tighter, more focused passage — but may lose surrounding context.
- **Larger chunks** (e.g., 1024–2048) preserve more context around each passage but may include irrelevant content in the retrieved segment.

A chunk size of **512–1024** is a reasonable starting point for most documentation. If your documents have dense, self-contained sections (e.g., API reference pages), lean toward smaller chunks. If they rely heavily on context across paragraphs (e.g., narrative procedures), lean larger.

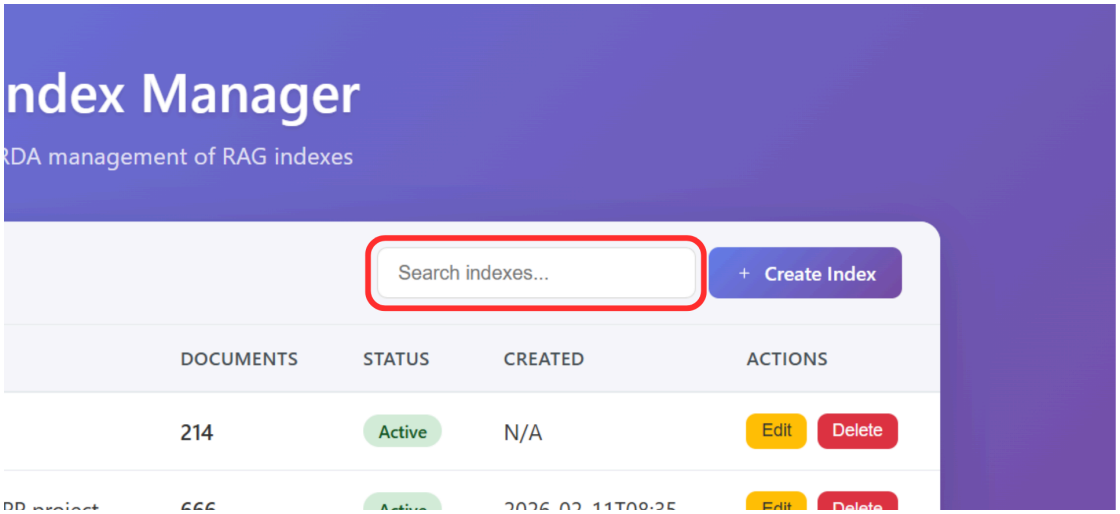
The Index Manager Dashboard

The Index Manager is the landing page at `http://10.55.55.1:7000`. It lists every index currently available.



Column	Description
Index Name	Unique identifier for the index in Azure AI Search. Click the name to open its edit page.
Description	A summary of the index's purpose.
Documents	Total number of processed document chunks in the index.
Status	Current state — **Active** means the index is reachable and ready to query.
Created	Timestamp of when the index was first created.
Actions	**Edit** to add content or update the description. **Delete** to permanently remove the index and all its data.

Use the **Search** bar at the top right to filter indexes by name.

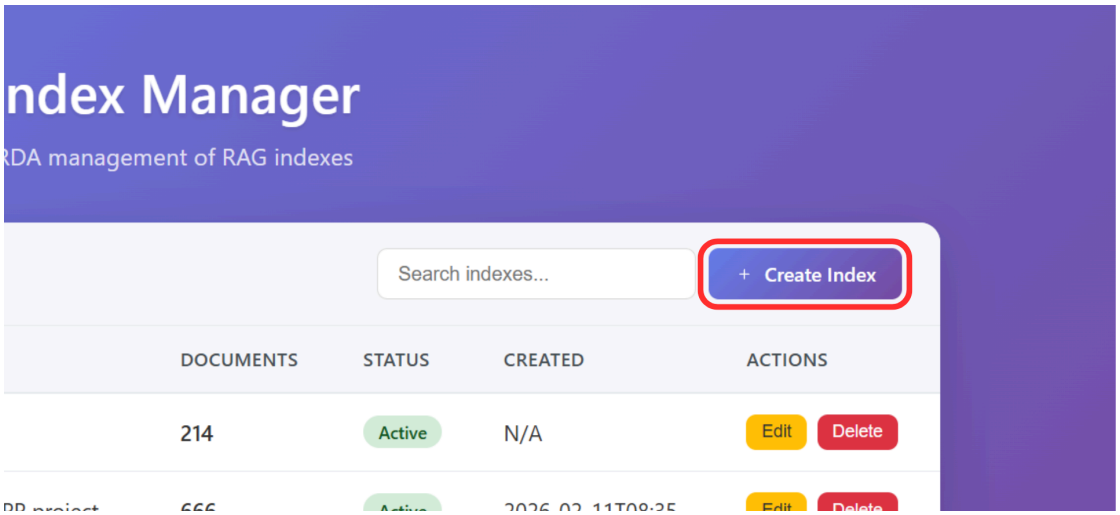


Warning

There is no way to view the individual files that have been added to an index — only the total document count is shown. Keep a separate record of what you have uploaded to each index if you need to track this.

Creating a New Index

Click **+ Create Index** on the dashboard to open the creation form.



Step 1: Configure the Index

Fill in the following fields:

- **Index Name** — A unique identifier. Use lowercase letters, numbers, and hyphens (e.g., `project-docs`). This name is used by the MCP server to reference the index as a tool.
- **Vector Dimensions** — Leave this at the default **3072** unless you are using a different embedding model.
- **Chunk Size** — Select from the dropdown. If unsure, start with **512** and adjust based on the quality of responses you get.
- **Description** — A short summary of the index's contents and purpose. This is for your own reference only — the AI model does not see this description when querying the index.

Step 2: Add Content

You can add content via file upload, a URL, or both at the same time.

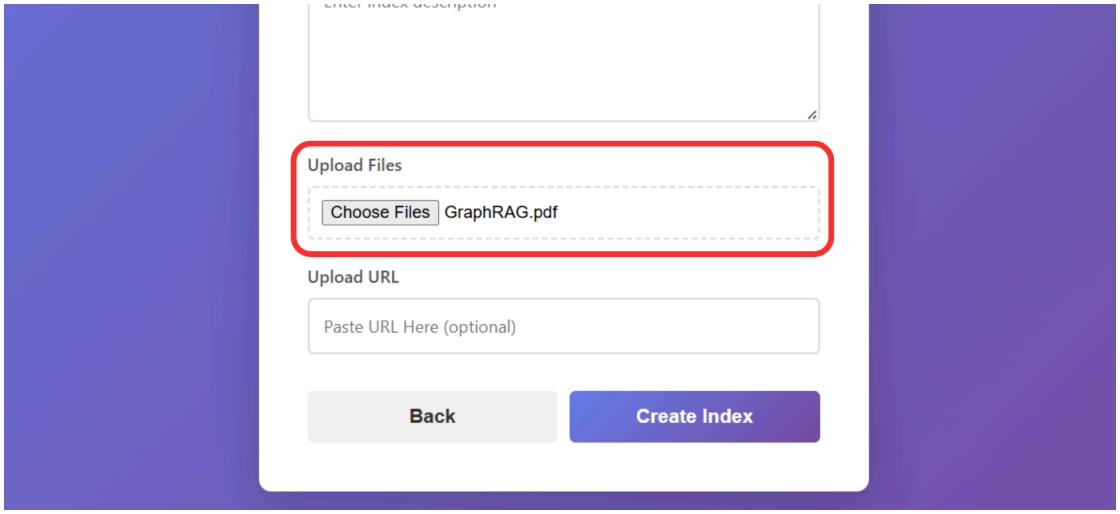
Upload Files

Click **Choose Files** and select one or more files from your computer. Supported types include:

- PDF documents
- Plain text files (`.txt`)
- Source code files (read as plain text)

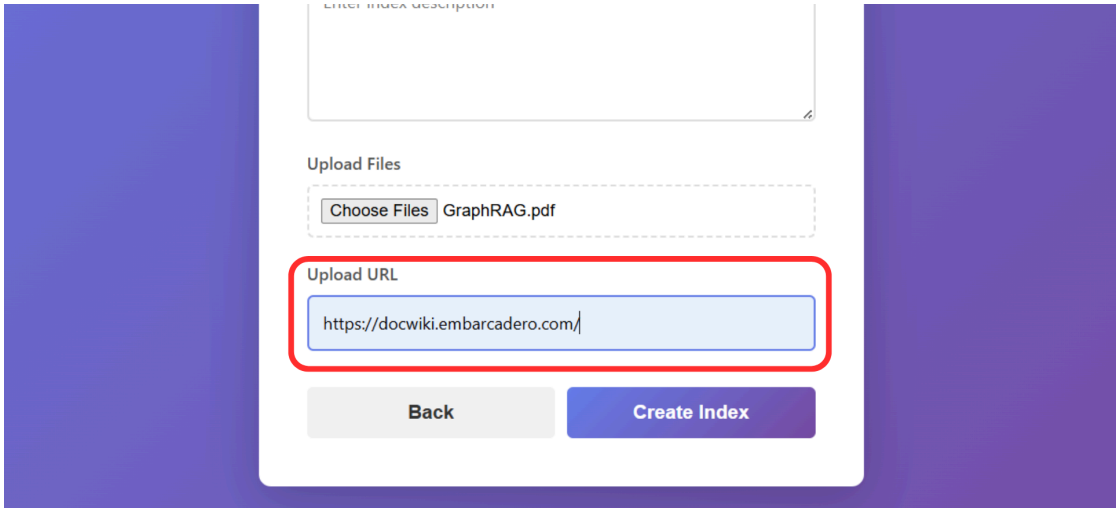
> **Warning**
>
> The index will accept all file types, but files with non-text content (e.g., images, binary files) will not produce meaningful results. Only upload files whose text content you want the AI to search.

> **Warning**
>
> The index does not prevent duplicate uploads. If you upload the same file twice, its content will be added to the index twice, which can cause the AI to return redundant results. Keep track of what you have already added.



Upload URL

Paste a URL into the **Upload URL** field and click **Create Index**. The site will scrape the page, find all links on that domain, and show you a **Select Links to Scrape** page. Check the boxes next to the pages you want to add to the index, then click **Add Selected to Index**.

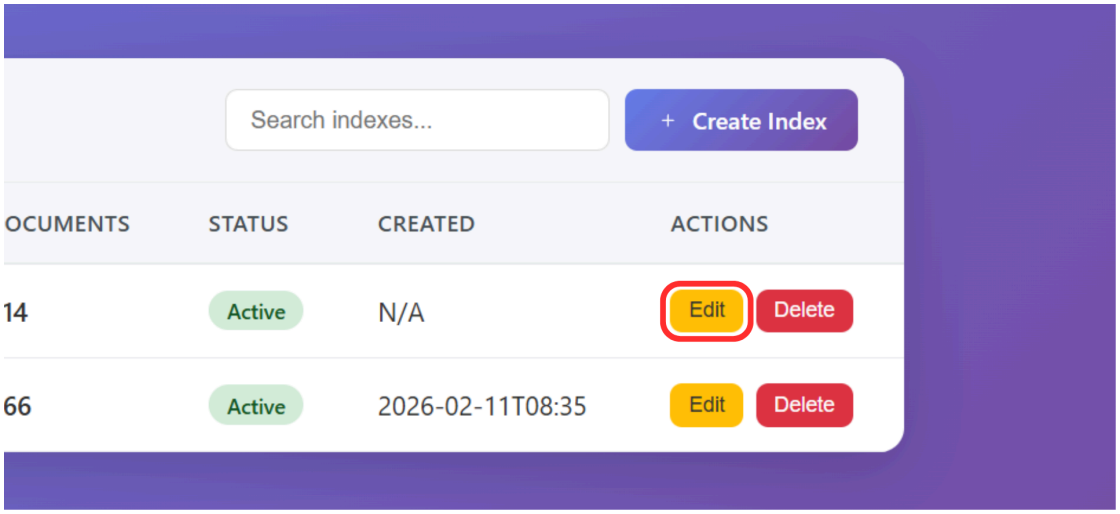


Step 3: Submit

Click **Create Index**. The system will process and embed your documents into Azure AI Search. Larger uploads take more time — the index will show as **Active** on the dashboard once indexing is complete.

Editing an Existing Index

Click **Edit** next to any index on the dashboard to open the Edit Index page.



Read-Only Information

These fields reflect the current state of the index and cannot be changed:

- **Created At** — When the index was first created.
- **Status** — Whether the index is currently active.
- **Number of Documents** — Total chunk count across all uploaded files.

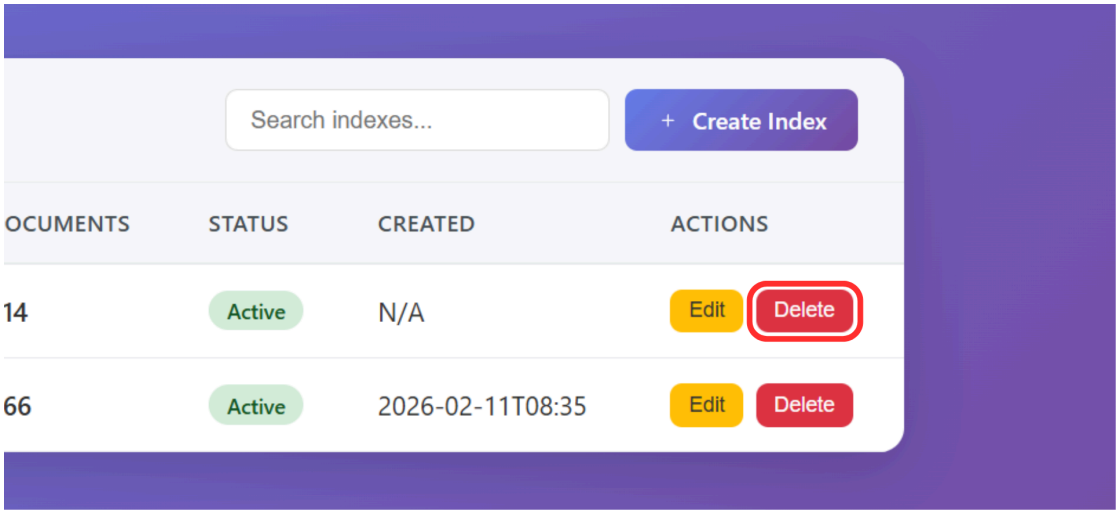
Updating the Index

- **Description** — Edit the text field to update the description for your own reference.
- **Add Files to Index** — Drag and drop files onto the upload area, or click **Choose Files** to browse. New files are added to the existing index without removing previously indexed content.
- **Add a Website URL to Scrape** — Enter a URL to scrape additional web pages into the index. This follows the same link-selection flow as during creation.

Click **Update Index & Add Files** to save changes and trigger ingestion for any new content.

Deleting an Index

Click **Delete** next to any index on the dashboard to permanently remove the index and all of its data from Azure AI Search.



> **Warning**
>
> Deleting an index cannot be undone. Make sure the index is no longer needed before removing it.

Using an Index with the AI

Indexes are exposed to the AI assistant as tools via the MCP server. To use an index during a conversation:

1. Connect your AI assistant (Cline or OpenWebUI) to the MCP server (see the CLine and OpenWebUI user guides).
2. In your AI assistant (Cline or OpenWebUI), the available indexes will be accessible tools for the LLM.
3. Ask a question — the AI will automatically query the relevant index and cite the passages it found.

> **Note**
>
> The AI does not have access to the index description. Give your indexes clear, consistent names so you can identify the right one when selecting tools.

Tips

- Keep each index focused on a single topic or project. Mixed-content indexes produce lower-quality answers.
- After adding new documents, allow a moment for indexing to complete before querying.
- Chunk size affects retrieval quality more than most other settings. If answers are missing context, try re-creating the index with a larger chunk size.

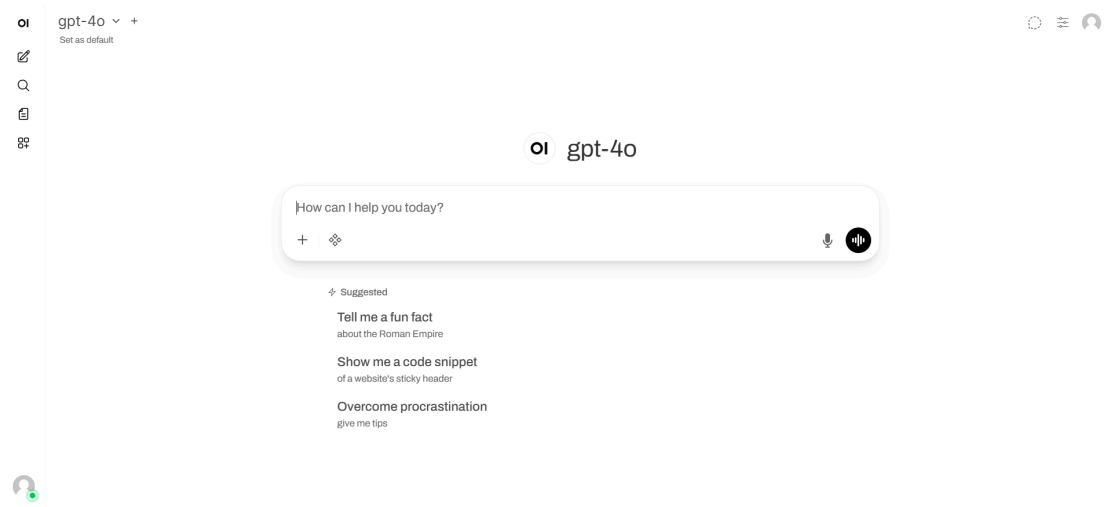
USER GUIDE

Open WebUI User Guide

This step-by-step guide will walk you through the basics of navigating the interface, selecting models, starting your first chat, and using the built-in workspace tools.

Access the WebUI

To get started, open your preferred web browser and navigate to the following address: `http://10.55.55.1:3000`

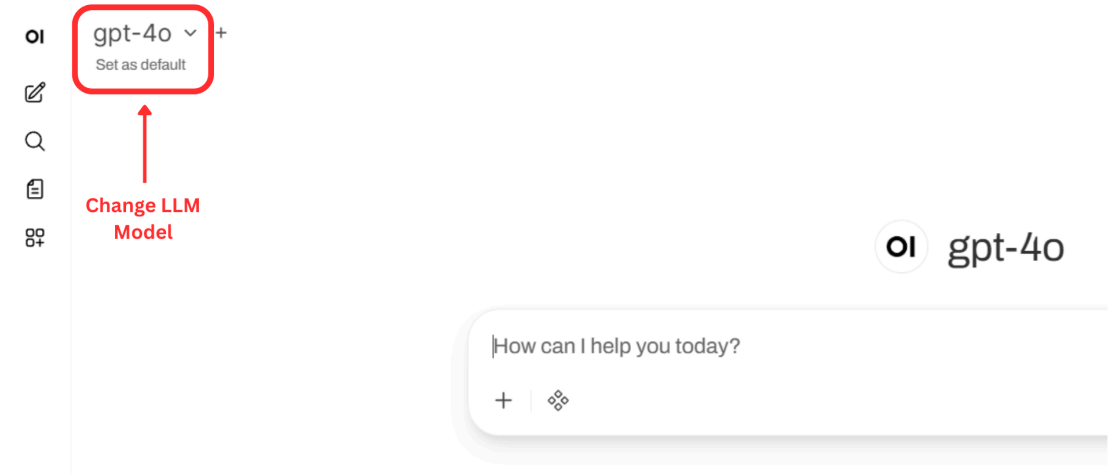


Choose Your LLM Model

Before you start chatting, you need to select the AI model you want to use.

1. Look at the top-left corner of the screen.
2. Click on the model dropdown menu (indicated by the red box in the image below).
3. Select your desired model (e.g., `gpt-4o`) from the list.

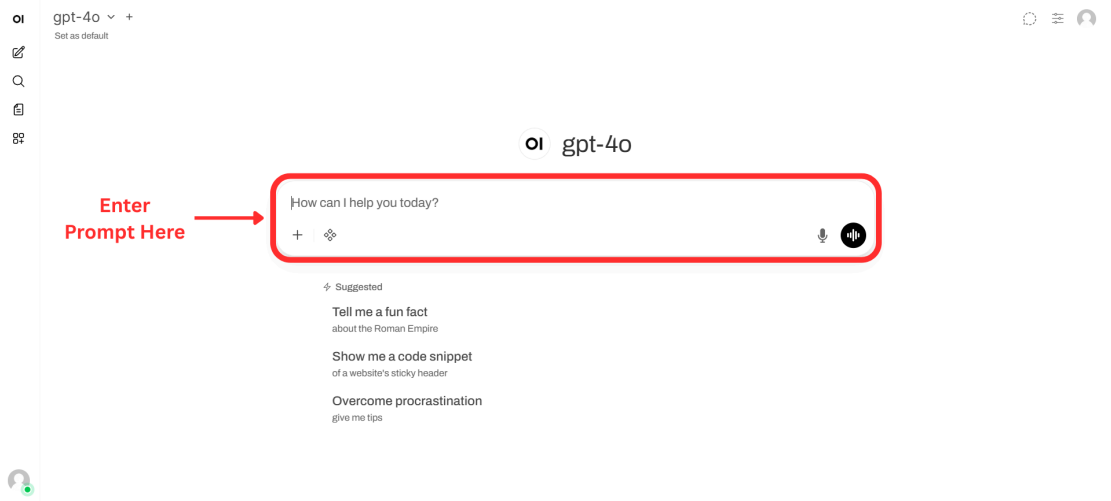
> **Tip:** You can click "Set as default" right below the model name so you don't have to choose it every time you log in.



Enter Your Prompt

Once your model is selected, you are ready to start generating responses.

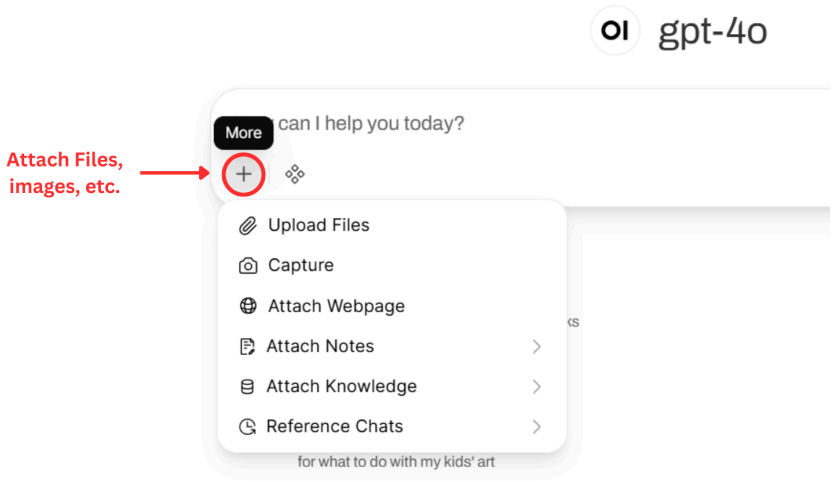
1. Locate the main chat box at the bottom of the screen.
2. Type your question, instructions, or prompt into this field.
3. Press **Enter** or click the send button to submit your prompt.



Include Files, Links, and More

If you need the AI to analyze a specific document, image, or website, you can easily attach it to your prompt.

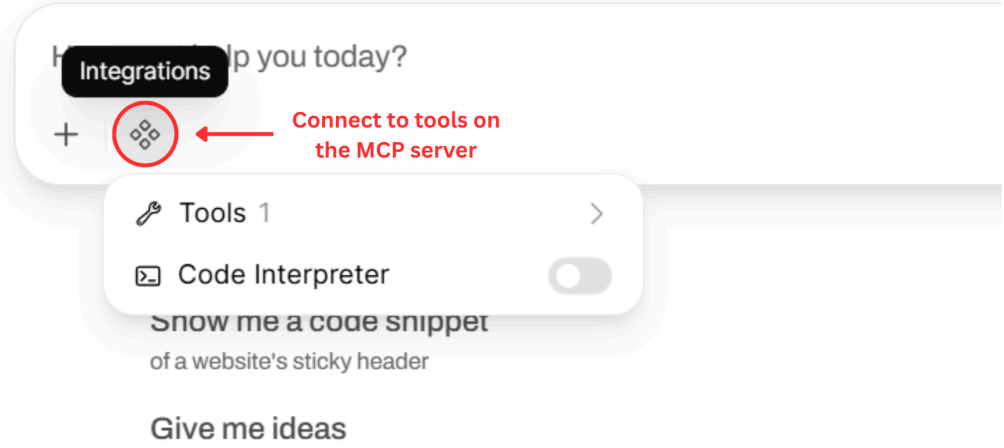
1. Click the `+` icon located on the left side of the chat input box.
2. A menu will appear with several options, including **Upload Files**, **Capture**, **Attach Webpage**, and more.
3. Select the appropriate option and choose the file or link you want to include.



Connect to MCP Server Tools

To use the custom search tools and retrieve information from the Azure AI Search indexes, you need to enable the MCP server integration.

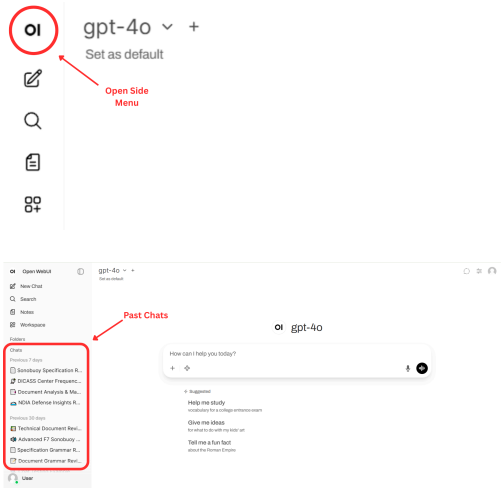
1. Locate the **Integrations** icon, shown as four small diamonds inside the chat input box, just to the right of the `+` button.
2. Click the icon to open the integrations menu.
3. Click **Tools** to view the available external tools.
4. Find **MCP Server** in the list and switch it to the **ON** position.
5. Once enabled, the LLM can autonomously call these tools to search the documentation while responding to your prompts.



Access Old Chats or Create a New One

Open WebUI automatically saves your previous conversations so you can pick up right where you left off.

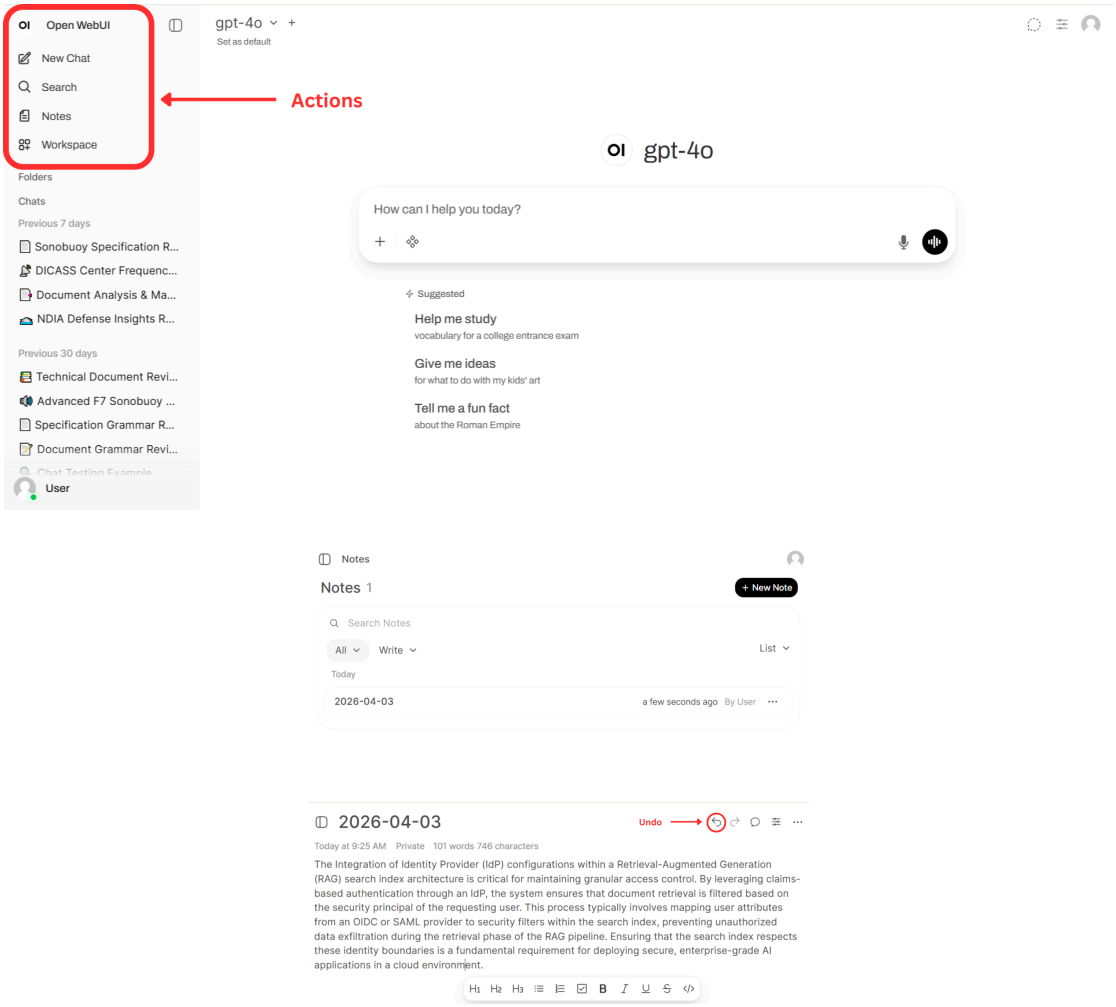
1. To open the sidebar, click the **OI** logo in the top-left corner of the screen.
2. The sidebar will slide out. From here, you can click **New Chat** at the top to start a fresh conversation.
3. To resume an old chat, browse your **Previous 7 days** or **Previous 30 days** history in the lower section of the menu.



Explore Additional Actions & Notes

The side menu also gives you access to a suite of other helpful workspace tools:

1. **Actions:** The upper section of the side menu contains quick links for **Search**, **Notes**, and your broader **Workspace**.
2. **Notes:** Clicking on "Notes" opens a dedicated interface where you can create, organize, and format your own documentation or save important AI outputs alongside your chats.



If you run into any issues or need further assistance, please reach out to the system administrator.